



Research article

Preparation and characterization of terephthalic acid from PET plastic waste via low-temperature alkaline hydrolysis

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ABSTRACT

This study presents a comprehensive investigation into the preparation and characterization of terephthalic acid from polyethylene terephthalate (PET) waste using low-temperature alkaline hydrolysis. The significance of this research lies in developing a sustainable and efficient method for recycling plastic waste, contributing to environmental pollution reduction, and promoting the circular economy concept. Low-temperature alkaline hydrolysis (80 °C) was employed to depolymerize PET into its basic units. Reaction conditions, including the molar ratio of (PET: NaOH), reaction time, and water to ethanol ratio in the solvent, were optimized to maximize the yield and quality of TPA. The final product was characterized using spectroscopic analysis techniques such as Fourier transform infrared spectroscopy (FTIR), Proton Nuclear Magnetic Resonance (¹H NMR) spectroscopy, and Gas Chromatography-Mass Spectrometry (GC-MS). Results demonstrated that the proposed method effectively produces high-purity terephthalic acid from PET waste with a yield of 99.5 % under the optimum conditions: (PET: NaOH) 1:4, water-to-ethanol ratio (20:80) at 80 °C, and a reaction time of 1 h. Moreover, the aqueous solution can be reused for multiple reaction cycles, enhancing process efficiency and reducing chemical and water consumption. This approach offers a sustainable PET recycling and resource recovery solution.

1. Introduction

Since the emergence of synthetic polymers in the early 20th century, industries have experienced radical transformation in their material use. Among these polymers, polyethylene terephthalate (PET) has emerged as one of the most significant engineering polymers owing to its unique physical and chemical properties. PET is characterized by high tensile strength and excellent chemical and thermal resistance, making it a fundamental material in the production of textile fibers, beverage containers, and plastic films [1, 2]. Chemically, PET is synthesized through a polymerization reaction between terephthalic acid (TPA) and ethylene glycol (EG) [3], both derivatives of crude oil. The polymer chain contains hydrolyzable ester bonds, which is a crucial characteristic in the recycling context. However, the stability of these bonds under normal conditions renders PET resistant to natural degradation, leading to its accumulation in the environment [4]. TPA is one of the most important chemicals for the production of PET and plasticizers [5]; With the depletion of non-renewable resources, including oil, coal, and natural gas, the production of TPA from organic wastes, especially waste PET, has gained significant attention. This shift in focus aligns with the principles of the circular economy and sustainable resource management [6]. With the global production of PET reaching approximately 50 million tons annually in 2014, coupled with an annual growth rate of 5 % [7], plastic waste management has become a significant environmental and economic challenge.

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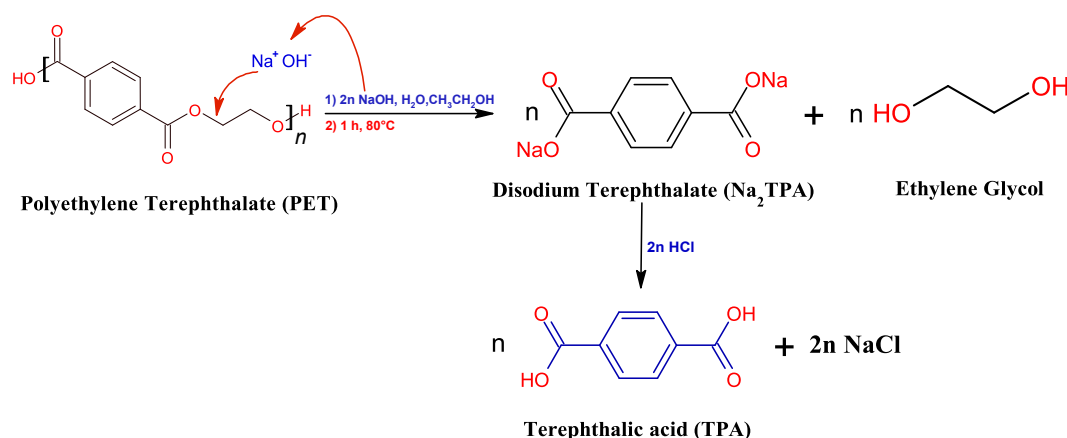


Fig. 1. The Chemical reaction of PET to TPA conversion.

Estimates suggest that by 2050, plastic waste could reach 33 billion tons, highlighting the urgent need to develop effective recycling strategies [8]. Historically, plastic waste disposal has primarily relied on direct incineration and landfilling. These practices lead to additional environmental problems, such as plastic leakage into the environment, microplastic decomposition inhibition, and the release of large quantities of greenhouse gases. With increasing environmental awareness and the adoption of sustainability concepts, the issue of plastic waste management has become a focus of global attention [9]. In response to this challenge, plastic waste management strategies have evolved to encompass three main axes: physical recovery [10,11], energy recovery [12], and chemical recovery [13–15]. Chemical recovery is considered the most promising approach, aiming to convert plastic waste into valuable chemical materials, thus realizing the concept of a circular economy in the plastics industry. Among chemical recovery methods, hydrolysis stands out as a promising technique for recycling PET [16,17]. This method is based on the cleavage of ester bonds in the presence of water, resulting in the formation of the original monomers: TPA and EG. However, traditional hydrolysis methods face significant challenges, typically requiring harsh conditions such as high temperatures and pressures, leading to substantial energy consumption and limiting their economic and environmental viability [18,19]. Consequently, the current research focuses on developing more efficient and environmentally friendly hydrolysis methods for PET. Recent trends include the use of co-solvent systems to enhance reaction efficiency and reduce reaction temperatures. These systems are believed to improve the wettability between the reaction solution and PET, thereby facilitating the degradation process [20].

In this context, the present research aims to develop an efficient and environmentally friendly method for PET degradation under moderate conditions using a high-concentration ethanol-water co-solvent system. This study focused on achieving high ester bond cleavage selectivity, improving the terephthalic acid recovery efficiency, and reducing the energy consumption and waste production. In addition to these core objectives, this study introduces a critical investigation into the reusability of the reaction solvent, aiming to significantly reduce chemical consumption and waste streams, thereby enhancing the overall sustainability and industrial relevance of the process.

This research represents a significant step toward achieving greater sustainability in the plastic industry by transforming PET waste into a valuable resource. Through the development of these technologies, we aim to contribute to the circular economy and achieve sustainable development goals while preserving natural resources and protecting the environment for future generations.

2. Experimental section

2.1. Materials

The materials used in this study were: Waste PET from used water bottles, Ethanol (C₂H₅OH), distilled water, hydrochloric acid (HCl), and sodium hydroxide (NaOH). All chemicals were of analytical reagent grade.

2.2. Sample preparation

For sample preparation, all plastic labels were removed from the mineral water bottles. The bottle surfaces were then thoroughly cleaned to ensure the removal of any residual adhesive, inks, or other printed materials. The samples were washed with distilled water and dried at room temperature. The PET bottles were cut into uniform small pieces (1 cm × 1 cm) using scissors. To ensure that the hydrolysis reaction occurs on the same surface area in all samples, the area to be cut must be a flat region. The PET bottles were then rinsed in deionized water and dried at room temperature.

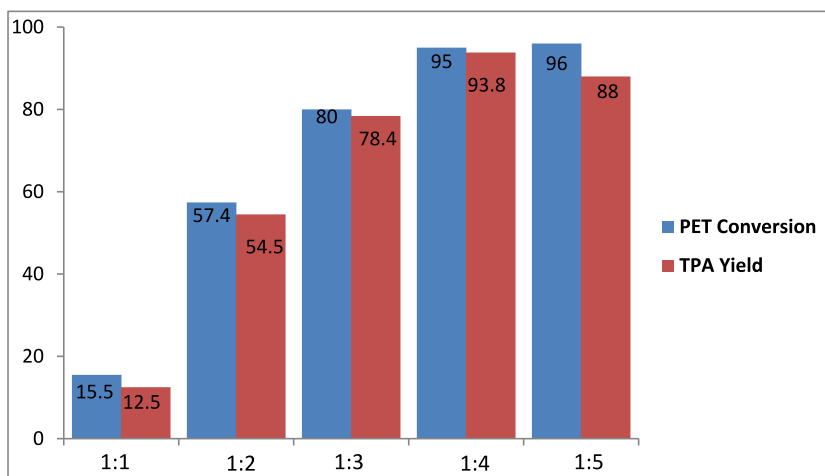


Fig. 2. Effect of the PET/NaOH molar ratio on PET conversion and TPA yield.

2.3. Hydrolysis process

The hydrolysis process consisted of two main stages. In the first stage, the hydrolysis reaction was conducted in a two-neck round-bottom flask connected to a condenser. Different molar ratios of (PET: NaOH) (1:1–1:5) were used. In a mixture of distilled water and ethanol with volume ratios ranging from 50:50 to 10:90. Polyethylene terephthalate (5 g) was added to the alkaline solution, and the mixture was heated to 80 °C with continuous stirring at 500 rpm. The reaction times were varied from 0.5 to 2.5 h.

2.4. Filtration and TPA recovery

In the second stage, upon reaction completion, the mixture is cooled to room temperature. Unreacted polyethylene terephthalate was separated from the solution via filtration. The unreacted residue was washed several times with distilled water and dried in an oven at 85 °C for 5 h. The first filtrate (remaining solution after filtration) containing Na₂-TPA and ethylene glycol was collected and separated by filtration. The Na₂-TPA salt was dissolved in a minimal amount of distilled water, followed by the gradual addition of hydrochloric acid with continuous stirring for 5 min until reaching a pH of approximately 2.6, resulting in terephthalic acid (TPA) precipitation. The white precipitate was filtered, washed several times with distilled water, and dried in an oven at 85 °C. The relevant chemical reactions are illustrated in Fig. 1. TPA was obtained with a yield of 99.5 % under the following optimum conditions: PET/NaOH(1:4), water to ethanol ratio (20:80), 80 °C, and reaction time of 1 h.

2.5. Calculations and analysis

The percentage of PET depolymerization was calculated using the following equation (1):

$$PET_{Depolymerization} = \left(\frac{W_{PET0} - W_{PET1}}{W_{PET0}} \right) \times 100\% \quad (1)$$

Where W_{PET0} represents the initial weight of PET, while W_{PET1} corresponds to the weight of unreacted PET [21]. The TPA yield was calculated according to the following equation (2):

$$TPA_{yield} = \left(\frac{W_{TPA1}}{W_{Theoretical\ TPA}} \right) \times 100 \quad (2)$$

$$TPA_{yield} = \left(\frac{W_{TPA1}}{\left(\frac{MW_{TPA}}{MW_{PET}} \right) \times (W_{PET0} - W_{PET1})} \right) \times 100\%$$

Where W_{TPA1} represents the weight of TPA produced, MW_{TPA} denotes the molecular weight of TPA (166.13 g/mol), and MW_{PET} correspond to the molecular weight of PET (192 g/mol) [22].

2.6. Characterization methods

The final product was characterized using several analytical techniques. Fourier Transform Infrared (FTIR) analysis was performed on a JASCO FTIR-460 Plus spectrometer using the KBr pellet method over a wavenumber range of 400–4000 cm⁻¹ to identify the

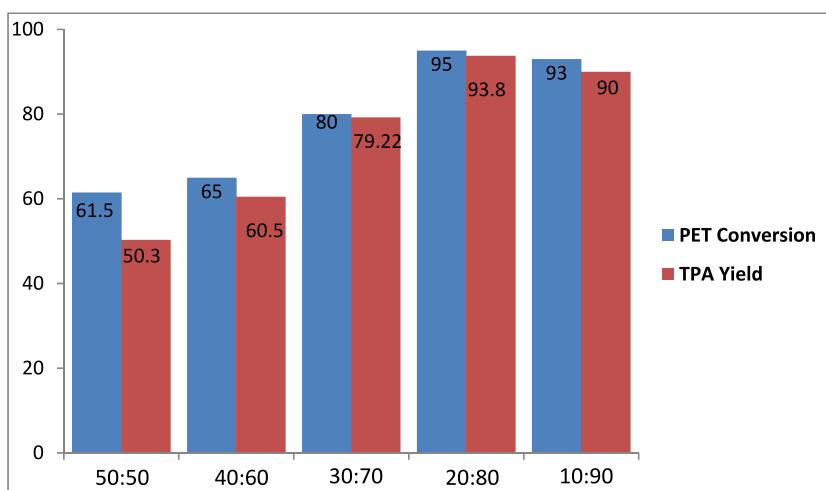


Fig. 3. Effect of water/Ethanol ratio on PET conversion and TPA yield.

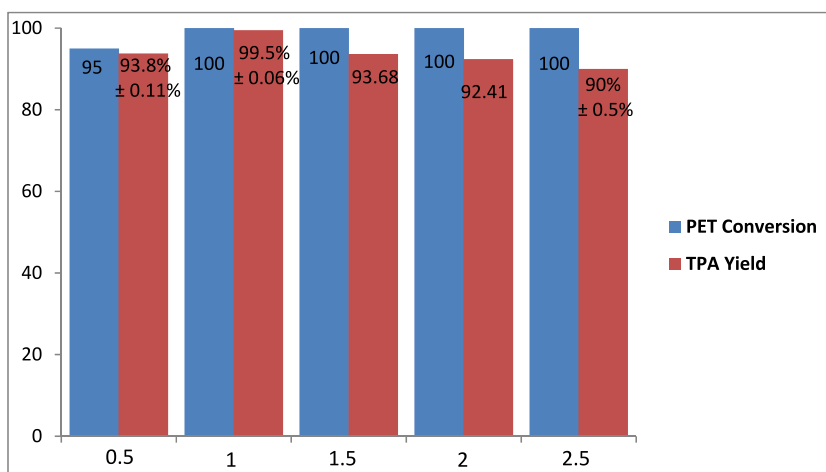


Fig. 4. Effect of reaction time on PET conversion and TPA yield.

functional groups. Proton Nuclear Magnetic Resonance (^1H NMR) spectra were recorded on a **Bruker Avance III 400 MHz** spectrometer, using deuterated dimethyl sulfoxide (DMSO-d_6) as the solvent. Gas Chromatography-Mass Spectrometry (GC-MS) analysis was carried out to confirm the purity of the product using an **Agilent-6890A GC system** coupled with an **Agilent-5975C Mass Spectrometric Detector**.

3. Results and discussion

3.1. Optimization of the reaction conditions

The effects of changing the reaction parameters such as the molar ratio of (PET: NaOH), reaction time, and water to ethanol ratio in the solvent on the hydrolysis efficiency and TPA production were investigated see (Figs. 2–4).

3.1.1. Effect of (PET/NaOH) molar ratio

The molar ratio of polyethylene terephthalate (PET) to sodium hydroxide (NaOH) is a critical factor in the PET depolymerization process. Ideally, 2 mol of NaOH are required to convert 1 mol of PET, as shown in the equilibrium reaction in Fig. 1. In this study, NaOH acts as a base, dissociating into hydroxide (OH^-) ions and sodium (Na^+) cations. The OH^- ions are nucleophilic and play a crucial role in breaking down the PET surface by attacking the carbon atoms in the carboxyl ester group (R-COO-R). The Na^+ cations then react with the terephthalate anions formed during this process to produce sodium terephthalate ($\text{Na}_2\text{-TPA}$). The sodium in $\text{Na}_2\text{-TPA}$ can be removed through acidification with hydrochloric acid (HCl), where the Na^+ reacts with the Cl^- ions from HCl, resulting in the formation of sodium chloride (NaCl), and terephthalic acid (TPA) [23]. To observe the effect of this factor, the depolymerization process

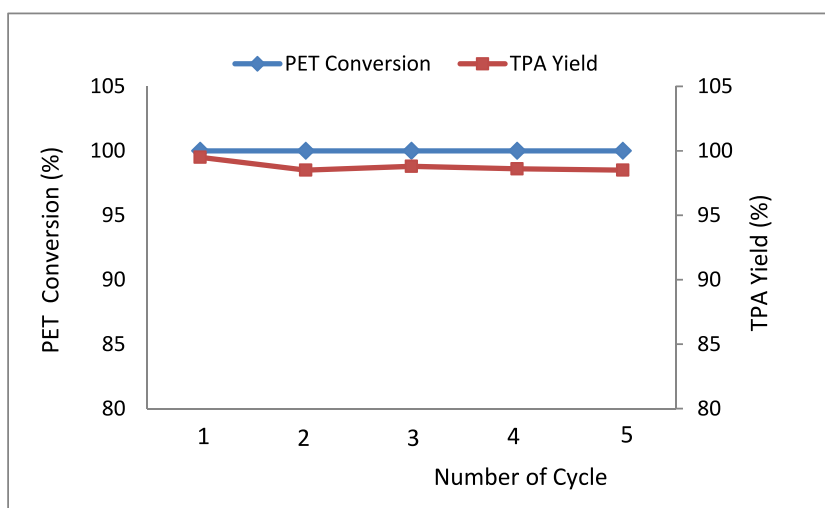


Fig. 5. Reuse of the reaction solution (water: ethanol(20:80), PET: NaOH(1:4)) for degradation of PET at 80 °C for 1 h in each cycle.

was carried out using a (water: ethanol) ratio of 20:80, 5 g of PET bottle, 50 mL total solution, a temperature of 80 °C, and a reaction time of 30 min. The amount of NaOH was varied between 1.04, 2.08, 3.12, 4.2, and 5.2 g, corresponding to (PET: NaOH) molar ratios of 1:1, 1:2, 1:3, 1:4, and 1:5, respectively. The results showed that the PET conversion efficiency also increased with increasing NaOH content. The conversion rate improved from 15.5 % to 95 % when the molar ratio was changed from 1:1 to 1:4 (Fig. 2). This improvement can be attributed to the increased reaction rate and degree of depolymerization due to the availability of more hydroxide ions, which enhance the hydrolysis process [24,25]. However, in the case of an excess of NaOH (1:5 M ratio), although the PET conversion improved, the production of TPA tended to decrease. This decline can be attributed to the deposition of excess NaOH on the PET surface, forming a barrier layer that hinders the accessibility of hydroxide ions to the reactive sites, as well as the high pH resulting from excess NaOH, which neutralizes the target product [26]. Therefore, a molar ratio of 1:4 was considered ideal, because it achieved a TPA yield of 93.8 %.

3.1.2. Effect of (water/ethanol) ratio

In this study, the effects of various water-ethanol mixture ratios on the depolymerization of polyethylene terephthalate (PET) and TPA yield were investigated. The experiments were conducted using a molar ratio (PET: NaOH) of (1:4) in a 50 mL solvent medium, and a reaction time of 30 min at 80 °C. The study examined a range of (water: ethanol) ratios, varying from (50:50) to (10:90). The effect of the (water: ethanol) ratio on PET conversion is shown in Fig. 3. Results demonstrated that the PET degradation rate is dependent on the mixture ratio of these two solvents [20,27]. The rate of PET degradation gradually increased with increasing ethanol content until a certain point was reached. The highest conversion rate was recorded at a (water: ethanol) ratio of 20:80, achieving a conversion efficiency of approximately 95 %. This can be attributed to the enhanced diffusion of hydroxide ions in the ethanol-water mixture, which improves the accessibility to PET ester bonds, coupled with the reduced polarity of the solvent system that facilitates the separation of terephthalate ions. It was noted that increasing the ethanol content beyond 80 % led to a decrease in conversion efficiency, confirming the importance of maintaining an adequate water content for optimal results. Similarly, the terephthalic acid (TPA) yield followed a comparable pattern, achieving the maximum yield (approximately 93.8 %) at a (water: ethanol) ratio of 20:80. Based on these findings, an optimal ratio of 20:80 (water: ethanol) was selected for subsequent experiments because this ratio achieved an ideal balance between conversion efficiency and TPA yield.

3.1.3. Effect of reaction time

The influence of reaction time on PET bottle (5 g) alkaline hydrolysis was investigated through experiments conducted at varying durations ranging from 0.5 to 2.5 h. The reaction conditions employed a (water: ethanol) ratio of 20:80 and a (PET: NaOH) ratio of 1:4 at 80 °C. The results are shown in Fig. 4. A significant correlation was observed between reaction time and PET conversion efficiency. The optimal reaction time of 1 h achieved the highest TPA yield (99.5 % ± 0.06 %) with minimal experimental error. The decrease in yield at 2.5 h (90 % ± 0.5 %) aligns with the expected effects of prolonged reaction times, such as side reactions and partial re-dissolution of TPA. This highlights the importance of optimizing the reaction time to maximize yield and minimize energy consumption. The degradation mechanism of PET through alkaline hydrolysis involves solid-liquid interactions, where PET particles undergo progressive size reduction until complete dissolution [28,29], resulting in monomer formation. During this process, PET is depolymerized into its fundamental components (Na₂-TPA and EG). Although extended reaction periods generally enhance the depolymerization efficiency [30], the identification of an optimal reaction time (1 h in this case) is crucial for balancing process efficiency, product yield, and economic considerations. This optimization is particularly important because longer reaction times require additional energy input to maintain the reaction conditions, potentially affecting operational costs and overall process sustainability.

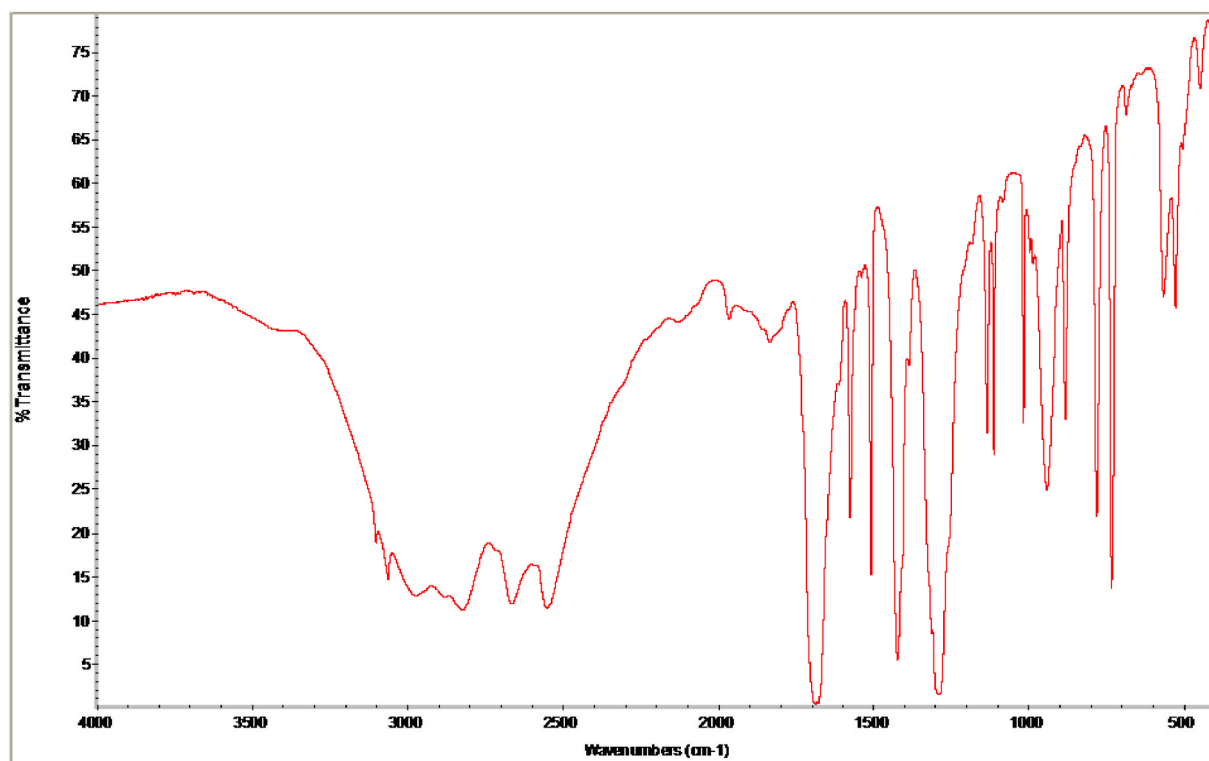
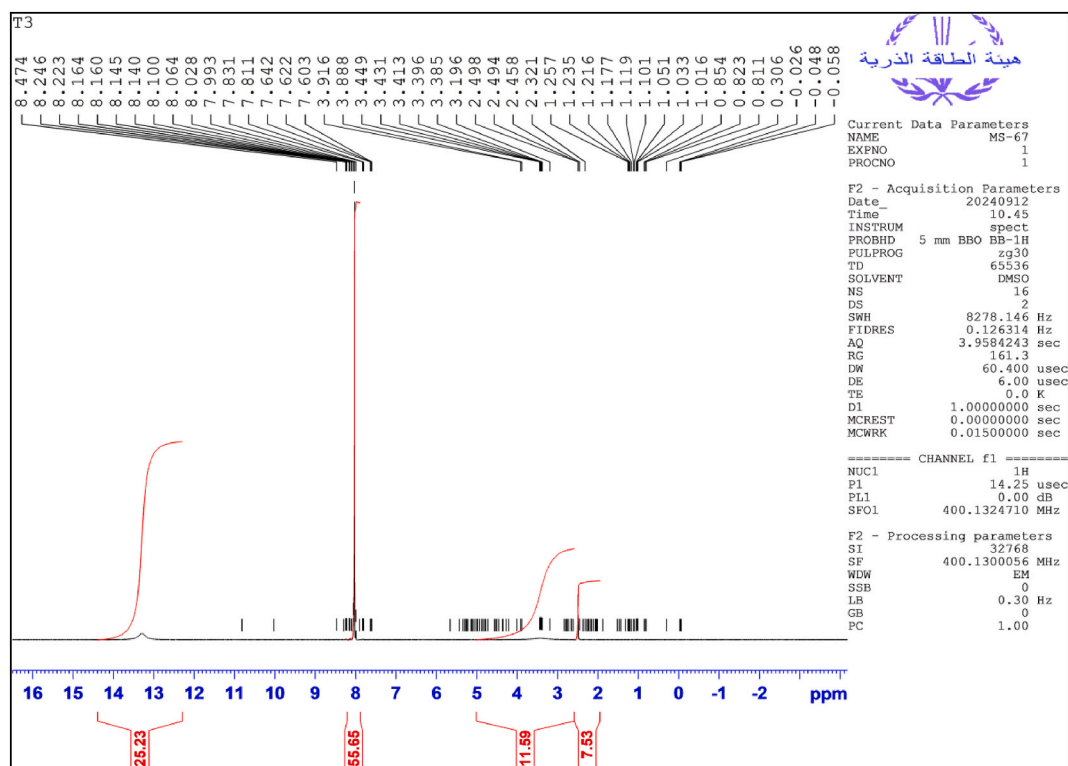


Fig. 6. FTIR spectra of TPA.

Fig. 7. ¹H NMR spectrum of the prepared TPA.

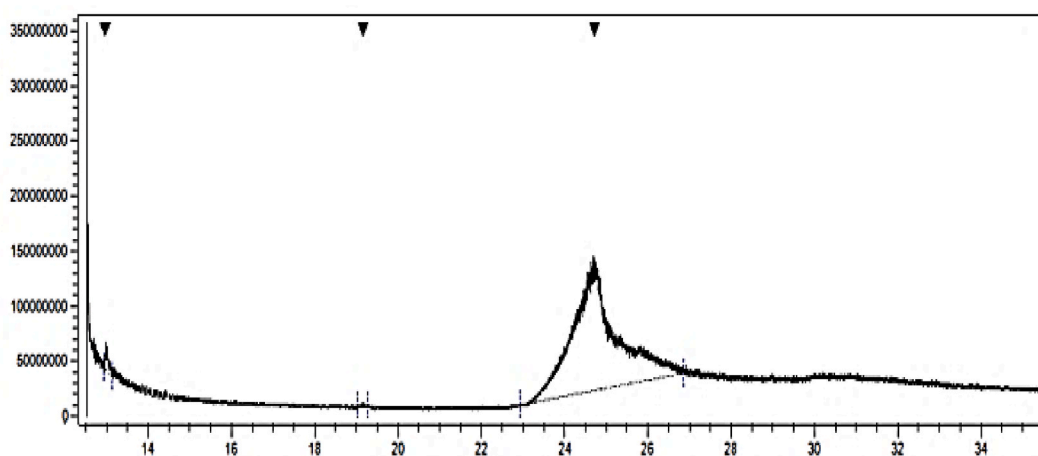


Fig. 8. GC-MS chromatograms of the prepared TPA.

[31]. The remarkably high conversion rate and TPA yield achieved at the 1 h mark suggest that these specific reaction conditions (20:80 (water: ethanol) ratio and 1:4 (PET: NaOH) ratio) create an optimal environment for PET depolymerization, achieving maximum efficiency while minimizing energy consumption and processing time.

3.2. Reuse of the reaction solution

A study was conducted to explore the reusability of the reaction solution. The methodology involved separating the ethanol-water-sodium hydroxide solution from the insoluble terephthalate, allowing its direct use in subsequent reactions. Due to the partial consumption of sodium hydroxide as a neutralizing agent, new NaOH was added to the reaction solution in the third cycle. The results demonstrate that the degradation efficiency maintains high levels of nearly 100 % throughout five consecutive cycles (Fig. 5). With TPA retaining its self-separation capability and stable production rates. The economic significance of this process is reflected in the substantial reduction in industrial production costs.

3.3. Characterization of the produced TPA

3.3.1. FTIR spectra

An analytical study was conducted to investigate the functional groups in TPA produced by alkaline hydrolysis of PET bottles. The analysis was performed using Fourier Transform Infrared Spectroscopy (FTIR). The obtained FTIR spectra were compared to the standard TPA spectrum. The results illustrated in Fig. 6 reveal distinctive spectral peaks confirming the presence of essential functional groups of TPA in the FTIR spectra, where the carboxyl group appeared in the range of $3200\text{--}2500\text{ cm}^{-1}$, a strong peak for the carbonyl group at 1695 cm^{-1} , and the carboxyl group at $1574\text{--}1424\text{ cm}^{-1}$. Additionally, specific vibrations related to aromatic dicarboxylic acid were observed at 3063.7 cm^{-1} for the (C-H) bond, 1695 cm^{-1} for the (C=O) bond, and 1280.5 cm^{-1} for the (C-O) bond, and vibrations specific to a 1,4-disubstituted benzene ring were observed at $450\text{--}520$, 1130.7 , and 1019 cm^{-1} . The FTIR results indicate that the produced TPA exhibits a high level of purity, as the absorption spectrum closely resembles that of the standard TPA [32], and the observed FTIR spectrum is similar to the reference sources [33,34].

3.3.2. ^1H NMR spectroscopy analysis

To further confirm the structure and purity of the prepared TPA, ^1H NMR spectroscopy was employed. The spectrum, shown in Fig. 7, exhibits two distinct signals characteristic of terephthalic acid (TPA) in a DMSO- d_6 solvent. A broad singlet is observed at 13.25 ppm, corresponding to the two acidic protons of the carboxyl groups (-COOH). A second singlet appears at 8.03 ppm, which is attributed to the four chemically equivalent aromatic protons on the benzene ring. The chemical shifts of these characteristic peaks are in strong agreement with previously published data for terephthalic acid [35–37]. Crucially, the integration of these peaks provides quantitative confirmation of the TPA structure. The raw integral values for the aromatic protons' signal (8.03 ppm) and the carboxyl protons' signal (13.25 ppm) were 55.65 and 25.23, respectively. Normalizing these values reveals a proton ratio of approximately 4:1.9. This calculated ratio is in excellent agreement with the theoretical 4:2 proton ratio expected for a pure TPA molecule. This quantitative evidence, combined with the precise chemical shifts, strongly corroborates the successful synthesis of high-purity terephthalic acid. The remaining signals at 2.50 ppm and 3.34 ppm correspond to the residual DMSO- d_6 solvent peak and trace water, respectively.

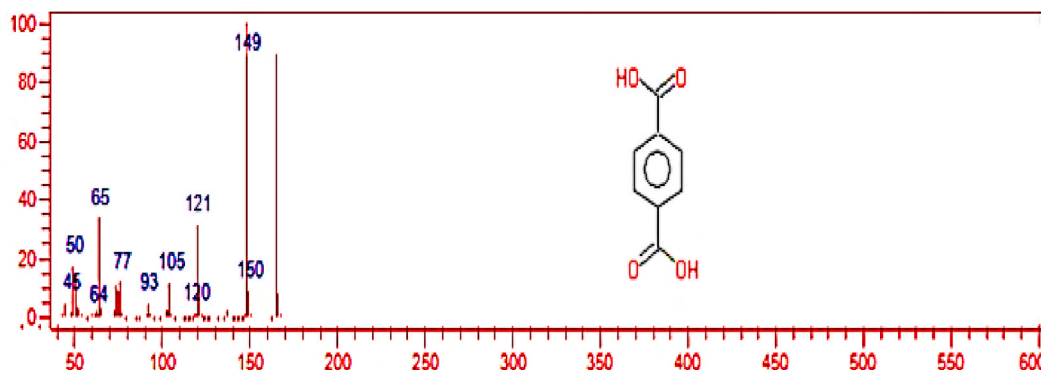
3.3.3. GC-MS analysis of TPA

Chromatographic analysis of the product extracted from polyethylene terephthalate (PET) depolymerization was performed using

Table 1

GC/MS analysis result of TPA prepared from PET bottles.

No	Compound	Molecular Formula	Molecular Weight	Retention Time(minutes)	Area %
1	Dimethyl Sulfoxide	C ₂ H ₆ SO	78.13	12,97	1.3
2	1-Ethanol, 2-(ethylsulfinyl)-	C ₄ H ₁₀ SO ₂	122.19	19,16	0.3
3	Terephthalic acid	C ₈ H ₆ O ₄	166.132	24,71	98,4

**Fig. 9.** Mass spectrum of TPA.**Table 2**

Review of the literature on alkaline hydrolysis of PET.

PET weight (g)	Solvent and Alkaline	Alkaline:PET molar ratio	Temperature (°C)	Reaction time (min)	PET Conversion (%)	TPA Yield (%)	Ref.
5	Distilled water:EtOH (20:80), NaOH (4.1 g)	4.1	80	60	100	99.5	This work
40	NaOH (20 g)	2.4	150	240	–	100	[38]
5.5	Ethylene glycol, NaOH (3 g)	2.7	200	25	–	92	[33]
10	Distilled water, NaOH (7 g)	7.10	99	150	85	–	[32]
30	Distilled water, NaOH (46.6 g)	3.46	100	240	34	–	[39]
15	Distilled water, KOH (42.7 g)	4	160	30	92.2	90.9	[40]
5	Ethylene glycol: Distilled water (50:50), NaOH (5 g)	4.95	90	35	92	–	[41]
10	Distilled water, NaOH (12 g)	5.76	100	240	>85	–	[42]
2	EtOH:Distilled water (60:40), NaOH (5 g)	12.37	80	20	95.23	–	[20]

Gas Chromatography-Mass Spectrometry (GC-MS). The analysis revealed three major compounds separated at retention times (Rt) of 12.97, 19.16, and 24.71 min, with relative peak areas of 1.4 %, 0.3 %, and 100 %, respectively Fig. 8. Table 1. The identity of the main compound (Rt = 24.71 min) was confirmed to be terephthalic acid (CAS: 100-21-0), with a 67.43 % match factor against the reference mass spectrum in the NIST database Fig. 9. The chromatogram shows high purity, with terephthalic acid representing most product. The minor compounds identified were dimethyl sulfoxide (DMSO) and alcohol derivatives, which may be residuals from the preparation process. These results confirm the effectiveness of the depolymerization process in producing high-purity terephthalic acid from polyethylene terephthalate (PET) bottles, with the purity reaching 98.3 % of the total product.

3.4. Comparison with other studies

A comparison with previous studies in the field of alkaline hydrolysis of polyethylene terephthalate demonstrates the superiority of our methodology. The process achieved complete PET conversion (100 %) and exceptional terephthalic acid yield (99.5 %) under moderate conditions, employing a molar ratio of 1:4 (PET: NaOH) and a water: ethanol ratio of 20:80 at 80 °C for 1 h. This methodology exhibits remarkable efficiency compared to previous studies Table 2, which typically require higher alkaline ratios, elevated temperatures, and extended reaction times to achieve similar results. The optimization of the alkaline concentration significantly reduced operational costs by minimizing the quantity of acidifying agents required during the isolation phase. Furthermore, the co-solvent system (water: ethanol 20:80) plays a crucial role in achieving these excellent results under moderate conditions, rendering this methodology more efficient and economically viable than conventional PET depolymerization approaches. This optimization represents a significant advancement in sustainable polymer recycling.

4. Conclusion

This research presents a significant advancement in polymer recycling technology by developing an efficient and sustainable approach for the depolymerization of polyethylene terephthalate (PET). The methodology employs alkaline hydrolysis under mild conditions at 80 °C using a water-ethanol co-solvent system that demonstrates remarkable effectiveness in waste PET conversion. The systematic optimization revealed that a water-ethanol ratio of 20:80, combined with a PET: NaOH molar ratio of 1:4, and a reaction time of 1 h, produced optimal results. Under these carefully tuned conditions, the process achieved an exceptional 99.5 % yield of terephthalic acid (TPA). The product's authenticity and high purity were conclusively verified through multiple analytical techniques, including FTIR, ¹H NMR, and GC-MS. The process distinguishes itself through several key innovations: the low operating temperature significantly reduces energy consumption; the co-solvent system facilitates efficient depolymerization; and most notably, the demonstration of the reaction solution's outstanding reusability across five consecutive cycles. This latter finding is of critical importance for enhancing the industrial viability and circularity of the process, collectively boosting its economic and environmental sustainability. This methodology's success in combining mild conditions, high yields, and solution recyclability establishes a valuable framework not only for PET recycling and for developing similar processes for other ester-containing polymers. This achievement represents a significant step toward more sustainable waste management practices, offering a practical solution that aligns with the principles of the circular economy while maintaining industrial feasibility.

CRedit authorship contribution statement

Mayas Saad: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Abdelkarim Elhamad:** Writing – review & editing, Validation, Supervision, Project administration, Data curation, Conceptualization. **Mohammad Deeb:** Writing – review & editing, Supervision, Data curation. **Hussam Alrakkad:** Writing – review & editing, Supervision, Conceptualization.

Data availability statement

All data generated or analyzed during this study are included within the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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